

Ruthenium

Ru (Ru I). Strongest classified transitions below 2 μm (NIST): 70 levels, 90 lines.

Ru + ionisation limit 59366.40 cm<sup>-1</sup> = 7.36050 eV (168.445 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element <[JferJ']>| in ea;  
Matrix elements are measured or NIST values; \* = high-accuracy theory. ∞ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

